MILITARY SPECIFICATION

PUMPING UNITS, HYDRAULIC MX-3196/APQ-83 AND MX-3197A/APQ-83

this specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be an explosion-proof hydraulic power supply either bench type or cart (portable) type, which will supply a minimum of 0.9 GPM of hydraulic fluid at 3000 PSI or 450 PSI ± 50 PSI

1.2 Classification - The Pumping Unit, Hydraulic (bench or cart) shall be of one type and shall consist of one of the following items.

Items	Type Designation	Appl. . Para.
Pumping Unit, Hydraulic	MX-3196/APQ-83	3.3
Pumping Unit, Hydaulic	MX-3197/APQ-83	3.3

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-H-5440	Hydraulic Systems: Design, Installation and Test of Aircraft (General Specification for)
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance
MIL-H-6083	Hydraulic Fluid, Petroleum Base, Preservative
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-H-8775	Hydraulic System Components Aircraft, General Specification for

FSC 4320

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ML-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Elect ronic and Fire Control Systems, General Specification for
MIL-R-21642	Radar Set AN/A PQ-83, Military Specification for
MIL-R-23373	Radar Set AN/APQ-94, Military Specification for

PUBL ICATIONS

SAR-300	Material Changes and Bulletins, Preparations of

Federal

FED-STD-No 595	Colors (collection of 358 standard colors for paints and other materials)

DRAWING

AND 10061	Tubing End, Standard Dimensions for Flared

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

2.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring acitivty.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3 REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200

(2) Nonrepairable subassemblies as outline in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity Nonrepairable subassemblies must be reliable.

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(3) When previously produced models of this equipment did not use nonrepairble subassemblies the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.2 Design and Construction -

3.2.1 General - This equipment shall fulfill the applicable design, workmanship and construction requirements of Specification MIL-T-21200, MX-3196/APQ-83 (Bench) to be Class 3 equipment and MX-3197A/APQ-83 to be Class 2 equipment, unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under the environmental and mechanical service conditions of Specification MIL-T-21200. Sand, dust, fungus, crash safety and immersion test shall not be applicable to this equipment.

3.2.2.1 Vibration - In accordance with Specification MIL-T-21200 paragraph 3.2.14.4 with these exceptions.

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 33	0.02 inch

A test at 0.02 inch Double Amplitude and the Rate of Frequency varying between 33 to 55 cycles per second is to be run for exploratory determination of mechanical transmissability. This test is to be non-destructive.

3.2.2.2 Shock - In accordance with Specification MIL-T-21200, paragraph 3.214.5.1 with the following exceptions.

(1) MX-3196/APQ-83 shall not suffer damage or subsequently fail to provide the performance specified m the detail equipment specification when subjected to 15 impact shocks of 15 g. consisting of 3 shocks in opposite directions along each of 2 horizontal and 1 vertical axis (not subjected to 3 drops on the opposite side of the vertical axis), each shock impulse having a time duration of 11 ± 1 milliseconds.

(2) MX-3197A/APQ-83 shall not suffer damage or subsequently fail to provide the performance specified in the detail equipment specification when subjected to 6 impact shocks of 15 g. consisting of 3 shocks on 1 vertical and 1 horizontal axis only (2 normally operating positions). each shock impulse having a time duration of 11 ± 1 milliseconds.

3.2.2.3 Humidity - In accordance with Specification MIL-E-5422.

3.2.2.4 Explosive Conditions - Hydraulic Pumping Unit MX-3197A/APQ-83 shall not cause ignition of an ambient explosive gaseous mixture with air when operating in such an atmosphere up to 10,000 feet in accordance with Specificat ion MIL-E-5422.

3.2.3 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests.

Temperature	Room Ambient (30° ±10°C)
Altitude	Normal Ground

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Vibration	None
Humidity	Room Ambient up to 90% relative humidity
Input voltage	115/200 VAC, 3 phase, 400 cps

3.2.4 Total Weight - The total weight of each unit, excluding cables, shall not exceed the weight limits shown in paragraph 3.3.1.2.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 45 minutes continuously or intermittently, without the necessity for readjustment of any controls.

3.2.6 Operating Life -

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requireing servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall *have* a minimum total operating life of 2000 hours with reasonable serv icing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.7 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-T-21200, Class 3, and shall give specified performance from the following power source with characteristics as defined hereinafter. The power required shall not exceed the specified amount.

(1)	AC Power (Three Phase)	
	Operating Voltage Limits	115/200 Volts ± 10 percent
	Operating Frequency Limits	380 to 420 CPS
	Power (3 Ø)	25 amperes (max) per phase, continuous rating
(2)	Motor 2-1/2 hp	Refer to paragraph 3.3.1.3.13(5) of this specification

3.2.8 Overload Protection. Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other devices shall be held to a minimum.

3.2.8.1 Under Voltage Protection - The equipment shall not be damaged by operation at 98 volts and shall automatically resume normal operation when the voltage returns within the specified operating limits.

3.2.9 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.10 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.11 Maintenance Provision and Field Testing - Provision for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.1.2 Interference Control - The generatino of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

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3.2.13 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200

3.3 Detail Requirements -

3.3.1 Pumping Unit, Hydraulic MX-3196/APQ-83 and Pumping Unit, Hydraulic MX-3197A/APQ-83 - Each unit shall meeet the following requirements.

3.3.1.1 Function - Lath unit shall supply the hydraulic power for the Antenna of Radar Sets AN/APQ-83 and AN/APQ-94 as specified in paragraph 1.1 f this specification.

3.3.1.2 Form Factor - The Pumping Unit MX-3196/APQ-83 (Bench) shall not exceed 20 by 20 by 26 inches. The weight shall not exceed 175 pounds, The Pumping Unit. Hydraulic MX-3197A/APQ-83 (Cart) shall not exceed 18-1/2 by 31 by 40-3/4 inches and the weight shall not exceed 210 pounds. The base frame of both units shall be of aluminum construction.

3.3.1.2.1 Mounting - Pumping Unit, Hydraulic MX-3196/APQ-83 shall be shelf mounted. Pumping Unit, Hydraulic MX-3197A/APQ-83 shall be mounted on a hand cart type two-wheel frame. This unit may be used setting vertically on its wheels, or horizontally setting on skids on the frame, shall have seven tie-down brackets, and shall have four brackets to prov ide for external storage of two 10-foot hydraulic hoses (furnished).

3.3.1.2.2 Finish - The Pumping Unit, Hydraulic MX-3196/APQ-83 ('Bench) shall be finished with gray enamel con forming to MIL-T-21200. The Pumping Unit Hydraulic MX-3197A/APQ-83 (Cart) shall be finished with No. 23538, Semi-gloss Flight Line Yellow, per Federal Standard No. 595.

3.3.1.3 Contents - Each pumping unit shall contain the following subassemblies necessary to operate the Antenna.

3.3.1.3.1 Pump - The hydraulic system shall deliver a minimum of 0.9 GPM at 3000 PSI or at 450 ± 50 PSI, obtained from a motor -driven pressure-compensated pump under standard conditions.

(1) The pump shall be coupled to the motor by means of a Drive-Type X Engine Accessory. Power shall be trasm itted through a splined shaft.

(2) The pump shall have a minimum capacity of O 9 GPM at 3000 PSI, with inlet pressures of 10 inches HG vacuum to 65 PSIG. Compensation shall begin at 3000 PSI and continue until the flow is reduced to O GPM at 3500 PSI maximum pressure.

(3) Pump design shall be commensurate with long life and reliability.

3.3.1.3.2 Bleed Valve - A bleed valve shall be installed in the pump retake lines to provide for the removal of any air that might enter the system during maintenance of the power supply.

3.3.1.3.3 Filter - Each unit shall be provided with a Military Standard high pressure filter with a removable MS28720 (-6) 10 micron element. In addition the Pumping Unit, Hydraulic MX-3197A/APQ-83 (Cart) shall also be provided with an additional filter (conforming to the above specification) mounted n the suction line.

3.3.1.3.4 Differential Pressure Switch and Warning Light - The Pumping Unit, Hydraulic MX-3197A/APQ-83 shall contain a differential pressure switch to–supply an electrical signal to a warning light, when the differential pressure is 10 to 15 PSI, to provide an indication of a clogged or dirty filter. The differential pressure switch shall be installed across the suction filter. The differential pressure switch and warning light shall be located in an explosion-proof electrical box.

3.3.1.3.5 Pressure Gage - Each unit shall be provided with a pressure gage (0 - 5, 000 PSI) which shall be panel mounted and may be connected with an AN approved flexible hose.

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3.3.1.3.6 Flow Rate Gage - Each unit shall be provided with a flow rate gate (0.2 to 1.5 GPM). The bench flowmeter shall be panel mounted, and the cart flowmeter shall be mounted on the side of the housing.

3.3.1.3.7 Three-Way Selection Valve and Low Pressure Relief Valve - A three -way selector valve shall be used in the press ure line to divert the fluid through a line having a relief valve set at 450 ± 50 PSIG. A choice of two pressures will thus be obtainable at the high pressure outlet 3000 PSI or 450 ± 50 PSI.

(1) The selector valve shall be three position, with a closed center to serve also as a shut-off valve. It shall be panel mounted.

(2) The low pressure relief valve shall have a fixed setting of 500 PSIG. Isolation from the 3000 PSI line shall be pro vided by a check valve.

33.1.308 Fittings - AN standard straight-threaded flow fittings shall be used throughout each system. The fittings shall be made of steel or aluminum.

3.3.1.3.9 Tubing - The high pressure tubing shall be made of stainless steel. The low pressure tubing shall be made of aluminum. The tubing shall be flared per AND10061.

3.3.1.3.10 External Outlets - One pressure outlet and one return outlet shall be provided. Each outlet shall consist of a self- sealing coupling half u ith dust cap and chain as follows:

(1) Pumping Unit, Hydraulic MX-3196/APQ-83 shall have self -sealing coupling half, Aeroquip No 015533-S4-4D(2) requested.

(2) Pumping Unit, Hydraulic MX-3197A/APQ-83 pressure line shall have 10 feet of hydraulic AN type hose with Aeroquip self -sealing coupling half with Union Nit No 306533 -S-10D The return line shall have 10 feet of hydraulic AN type hose with Aeroquip self -sealing coupling half wiht Union Nut No 014533 -S5-16D

3 3 1 3. 11 Reservior - The reservoir for Pumping Un it, Hydraulic MX-3196/APQ-83 shall have a capacity of 4 5 gallons, and shall be made of aluminum The reservoir shall be equipped with an inspection cover and gasket, fill strainer, drain valve Schroeder valve (to provide for pressurization of approximately 3 PSIG). and a fluid level indicator marked to indicate high and low fluid levels The Pumping Unit, Hydraulic MX-3197A/APQ-83 shall not include a reservoir

3.3.1.3.12 Accumulator - The hydraulic pressure accumulator for MX-3197A/APQ-83 shall be the thermal expansion type.

3.3.1.3.13 Electrical System - The electrical control system shall meet the following requirements:

(1) Each AC line shall be protected against overload by automatic reset type relays.

(2) A thermoswitch shall be provided in the pump bleed line to shut off the motor if the fluid temperature reaches 200° ±5

(3) The magnetic contactor shall be a three-pole contactor.

(4) The magnetic contactor, overload relays, and thermoswitch shall be housed in a single explosion-proof housing.

(5) The pump motor shall be a 2-1/2 HP, 115/200 VAC, 3 phase, 4 wire, 400 cycle, explosion-proof motor.

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{6) Power shall be supplied to the Pumping Unit, Hydraulic MX-3196/APQ-83 by means of an MS-type connector, to mate with MS3106A-22-23S. Power shall be supplied to the Pumping Unit, Hydraulic MX-3197A/APQ-83 by means of a receptacle on the side of the unit to mate with an AN3430 connector. Power cables for each unit shall not be furnished by the manufacturer.

3.4 Performance - Unless otherwise specified values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5 Preproduction Sample - Unless otherwise specified in the contract or order, a preproduction sample of the Pumping Units, Hydraulic MX-3196/APQ-83 and MX-3197A/APQ-83 shall be manufactured such that it is representative of the production lot in all respects. The sample will be tested as specified in section 4 herein and is for the purpose of determining that the contractor prior to his starting production is capable of yeilding items that comply with the technical requirements of the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Tests Required - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests

(2) Sample Tests

(3) Acceptance Test

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the-production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with MIL-T-5422 to the extent

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specified herein, interference tests in accordance with MIL-I-6181 and drop tests for transit and combination cases in accordance with Specification MIL-T-21200.

4.2.3 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall by delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract.

4.3 Sample Tests - When requested by the procuring activity one equipment shall be selected from the first 10 production equipments submitted for acceptance and shall have successfully passed the individual tests.

4.3.1 Scope of Test - As a minimum, each equipment selected for sampling tests shall be subjected to the following test:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

4.3.2 Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity.. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease unitl proper corrective action is approved and accomplished. Corrective action shall also be accomplished or equipment previously accepted when requested by the procuring activity.

4.4 Acceptance Tests. - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall by signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of teh acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Special Tests

(3) Life Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equpiment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test. - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure satisfactory equipment operation.

4 4 1 3 Manufacturing Reliability Test - Each equipment shall be operated at the conditions specified herein a total accrued running time of 6 hours Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment

Temperature	Ambient Room
Humidity	Ambient Room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a maximum double amplitude of 0.02 inches.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes as specified herein The direction of vibration should be vertical to the normal mounting plane The equipment shall be non-operating during the 10 minute vibration period The 6-hour period of operation shall include that time necessary for performing all equipment tests and calibrations Only that time necessary to make a total accrued running time of 6 hours is required for operation after the 10-minute vibration period During the 6-hour period of operation the equipment shall be mechanically cycled periodically through its various phases of operation Should a failure occur, it should be repaired and the test continued Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment A record shall be kept of all failures

4.4.2 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.2.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run

(3) On an early equipment after an engineering change

(4) Whenever failure reports or other information indicate additional equipments should be tested (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance my require variations to assure satisfactory control.

4.4.2.2 Scope of Tests - Special tests shall consit of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum the following tests shall be made, except for those equpiments selected as a result of design changegs and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration 'Test (60 minute cycling in each of 3 planes)

(3) Humidity Tests

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4.4.3 Life Tests - When requested by the procuring activity, one unit shall be selected from current production equipment and shall have successfully passed the Individual Test.

4.4.3.1 Scope of Test - A 200 hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.4.3.1.1. A performance check shall be made to determine the equipment is operating satisfactorily.

4.4.3.1.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal Room
Altitude	Normal Ground (0 - 5000 ft)
Humidity	Room Ambient
A C Voltage	115 ± 5 volts (at lowest applicable frequency)
D c Voltage	27 0 ± 2 0 volts

4.4.3.1.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3.1.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.4.3.2 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.4.3.3 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval proposed corrective action intended to reduce the posssibility of the same failure(s) occuring in future tests.

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(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptante tests and life tests shall be prepared by the contractor and submitted to the procuring activity for I review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparatino of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.6 Presubmissino Testing - No item. part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Equipment Changes. - After approval of the preproduction model, no changes which affect weight installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by teh procuring activity. To obtain procuring activity approval the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary - sary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals Material Changes, and Material Bulletins shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract in accordance with Specification MIL-P-7936. All hydraulic units shall be flushed with type MIL-H-6083 preservative.

6. NOTES

6.1 Test Values - Normal and limiting valuees of performance data shall be determined at input voltage 115 ± 10 VAC These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 The parentheses (*), or type designation used throughout this specification shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.9. The complete type number shall be used on nameplates. shipping records, and instructor books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded be implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that mau in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO 45 R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION (No. and abbrev. title)

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1 HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4 REMARKS (Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)

SUBMITTED BY (Print name and title)	DATE

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D C

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